



POLITECNICO
MILANO 1863

Dynamics of Electrical Machines and Drives

Induction Motor model and R-FOC control

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Induction Motor Modeling

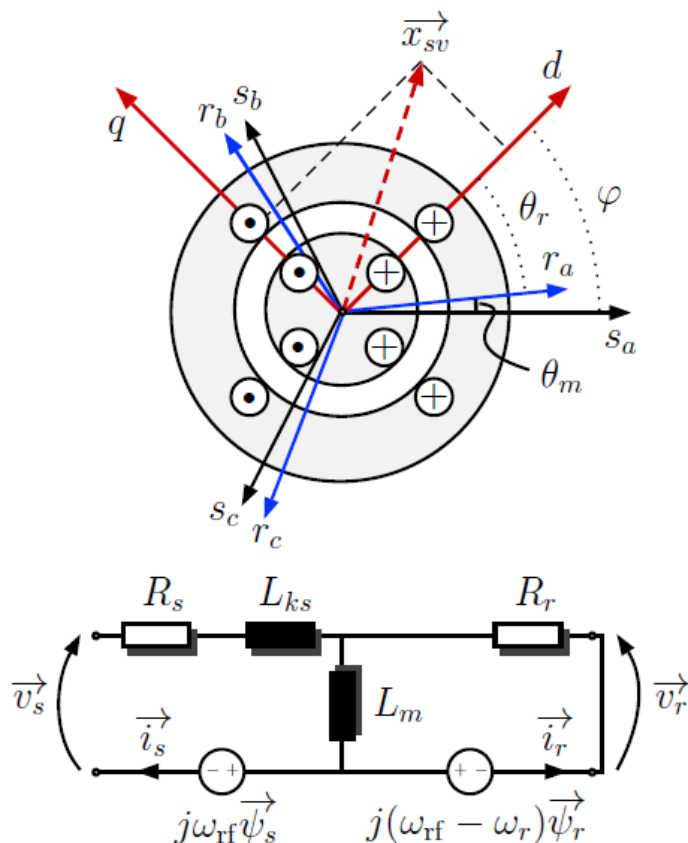
Approach

- The **modeling process** of induction machines is typically done in two/three steps:
 - ① starting from three-phase *abc* quantities and fundamental physical laws, such as Faraday's law of induction and Lorentz force, we derive machine's differential equations
 - ② transform the model into an orthogonal reference frame to simplify the representation, here we can use
 - **space vector notation**
 - **matrix notation**
 - state-space notation
 - ③ sometimes the machine model is translated from SI to normalized quantities (e.g. per unit)

(The differential equations derivation is beyond the scope of this lesson)

Induction Motor

- The dynamic model of a three-phase IM with *orthogonal* axes dq is:



$$\Gamma\text{-model} \rightarrow L_{ks} = L_s + L_m^2/L_r$$

IM motor model (sv form)

$$\vec{v}_s = R_s \vec{i}_s + p \vec{\psi}_s + j \omega_{\text{rf}} \vec{\psi}_s$$

$$\vec{v}_r = R_r \vec{i}_r + p \vec{\psi}_r + j (\omega_{\text{rf}} - \omega_r) \vec{\psi}_r$$

$$\vec{\psi}_s = L_{ks} \vec{i}_s + \vec{\psi}_r$$

$$\overrightarrow{\psi}_r = L_m(\overrightarrow{i}_s + \overrightarrow{i}_r)$$

$$m_e = n_p \Im \left(\overrightarrow{i_s} \underline{\psi_s} \right)$$

arbitrary angular speed ω_{rf} 

- **slip:** $s = 1 - \frac{\omega_r}{\omega_s} = 1 - \frac{n_p \omega_m}{\omega_s}$

Space Vector Orientation

Simulink Implementation

Considering the Park and Clarke transformations:

① Space Vector in Stationary Reference Frame

- $(\alpha\beta) \rightarrow \varphi = 0, \omega_{rf} = 0$
 $\vec{x}_{\alpha\beta} = \sqrt{\frac{2}{3}} (x_a + ax_b + a^2 x_c) = x_\alpha + jx_\beta \quad a = e^{j\frac{2\pi}{3}} \quad a^2 = e^{j\frac{4\pi}{3}}$

② Space Vector in Rotating Reference Frame

- $(dq) \rightarrow \varphi$ varying (we can select any rotating speed)
 $\vec{x} = \vec{x}_{\alpha\beta} e^{j\varphi} = \left[\sqrt{\frac{2}{3}} (x_a + ax_b + a^2 x_c) \right] e^{j\varphi} = (x_\alpha + jx_\beta) e^{j\varphi} = x_d + jx_q$

the angle is $\varphi = \int \omega_{rf} dt$ while $\vec{x} = \vec{v}_s, \vec{v}_r, \vec{i}_s, \vec{i}_r, \vec{\psi}_s, \vec{\psi}_r$

Model in Matrix Notation

Induction Motor

- The dynamic model of a three-phase IM with *orthogonal* axes dq is:

IM motor model (matrix form)

$$\mathbf{v}_s = R_s \mathbf{i}_s + p \boldsymbol{\psi}_s + \omega_{rf} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \boldsymbol{\psi}_s$$

$$\mathbf{v}_r = R_r \mathbf{i}_r + p \boldsymbol{\psi}_r + (\omega_{rf} - \omega_r) \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \boldsymbol{\psi}_r$$

$$\boldsymbol{\psi}_r = L_{ks} \mathbf{i}_s + \boldsymbol{\psi}_r$$

$$\boldsymbol{\psi}_s = L_m (\mathbf{i}_s + \mathbf{i}_r)$$

$$m_e = n_p (\boldsymbol{\psi}_s \times \mathbf{i}_s) = n_p (\psi_{sd} i_{sq} - \psi_{sq} i_{sd})$$

- remember that for a general variables $\mathbf{x} = \begin{bmatrix} x_d \\ x_q \end{bmatrix} = \begin{bmatrix} \Re(\vec{x}) \\ \Im(\vec{x}) \end{bmatrix}$
- it is valid for $\mathbf{x} = \mathbf{v}_s, \mathbf{v}_r, \mathbf{i}_s, \mathbf{i}_r, \boldsymbol{\psi}_s, \boldsymbol{\psi}_r \rightarrow \mathbf{v}_s = \begin{bmatrix} v_{sd} \\ v_{sq} \end{bmatrix} = \begin{bmatrix} \Re(\vec{v}_s) \\ \Im(\vec{v}_s) \end{bmatrix}$

Simulink Model in Space Vector Notation

Stationary alfa-beta model

- **Electrical part:** (squirrel cage $\rightarrow \vec{v}_r = 0$)

Stationary $\alpha\beta$ model

$$\vec{v}_{s\alpha\beta} = R_s \vec{i}_{s\alpha\beta} + p \vec{\psi}_{s\alpha\beta}$$

$$0 = R_r \vec{i}_{r\alpha\beta} + p \vec{\psi}_{r\alpha\beta} - j \omega_r \vec{\psi}_{r\alpha\beta}$$

$$\vec{\psi}_{s\alpha\beta} = L_{ks} \vec{i}_{s\alpha\beta} + \vec{\psi}_{r\alpha\beta}$$

$$\vec{\psi}_{r\alpha\beta} = L_m (\vec{i}_{s\alpha\beta} + \vec{i}_{r\alpha\beta})$$

$$m_e = n_p \Im \left(\vec{i}_{s\alpha\beta} \underline{\psi}_{s\alpha\beta} \right)$$

Classical form

$$\vec{\psi}_{s\alpha\beta} = \int (\vec{v}_{s\alpha\beta} - R_s \vec{i}_{s\alpha\beta}) dt$$

$$\vec{\psi}_{r\alpha\beta} = \int (-R_r \vec{i}_{r\alpha\beta} + j \omega_r \vec{\psi}_{r\alpha\beta}) dt$$

$$\vec{i}_{s\alpha\beta} = \frac{1}{L_{ks}} (\vec{\psi}_{s\alpha\beta} - \vec{\psi}_{r\alpha\beta})$$

$$\vec{i}_{r\alpha\beta} = \frac{\vec{\psi}_{r\alpha\beta}}{L_m} - \vec{i}_{s\alpha\beta}$$

$$m_e = n_p \Im \left(\vec{i}_{s\alpha\beta} \underline{\psi}_{s\alpha\beta} \right)$$

- **Mechanical part:**

$$J \frac{d\omega_m}{dt} + \beta \omega_m = m_e - m_l \quad \text{where} \quad \omega_r = n_p \omega_m$$

- Remember that j can be rewritten as $e^{j\frac{\pi}{2}} = \cos\frac{\pi}{2} + j\sin\frac{\pi}{2} = j$

Simulink Model in Space Vector Notation

Stationary alfa-beta model

Classical form (Complete)

$$\vec{\Psi}_{s\alpha\beta} = \int (\vec{v}_{s\alpha\beta} - R_s \vec{i}_{s\alpha\beta}) dt$$

$$\vec{\Psi}_{r\alpha\beta} = \int (-R_r \vec{i}_{r\alpha\beta} + j\omega_r \vec{\Psi}_{r\alpha\beta}) dt$$

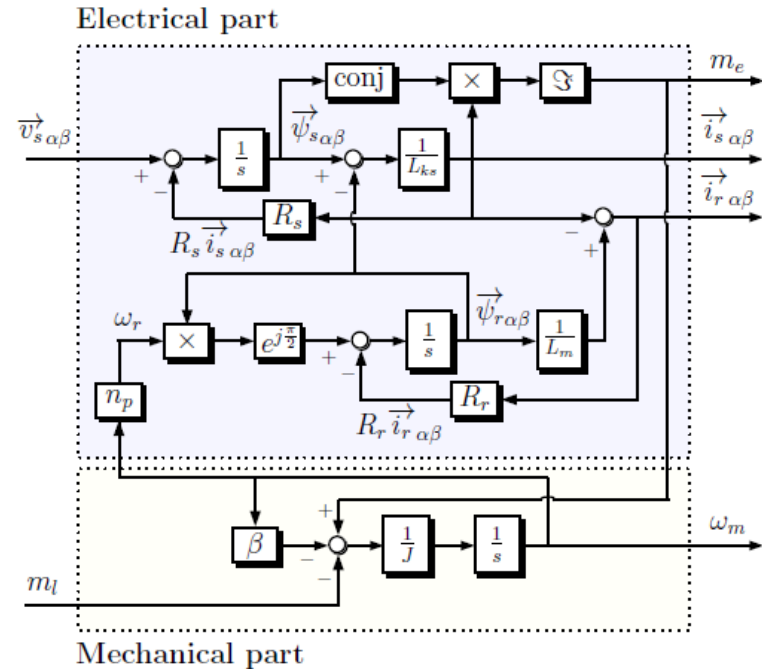
$$\vec{i}_{s\alpha\beta} = \frac{1}{L_{ks}} (\vec{\Psi}_{s\alpha\beta} - \vec{\Psi}_{r\alpha\beta})$$

$$\vec{i}_{r\alpha\beta} = \frac{\vec{\Psi}_{r\alpha\beta}}{L_m} - \vec{i}_{s\alpha\beta}$$

$$m_e = n_p \Im (\vec{i}_{s\alpha\beta} \vec{\Psi}_{s\alpha\beta})$$

$$\omega_m = \int \frac{1}{J} (m_e - m_l - \beta \omega_m) dt$$

- input: $\vec{v}_{s\alpha\beta}, m_l$
- output: $\vec{i}_{s\alpha\beta}, \vec{i}_{r\alpha\beta}, m_e$



→ Γ-model (space vector)

Brief Exercise

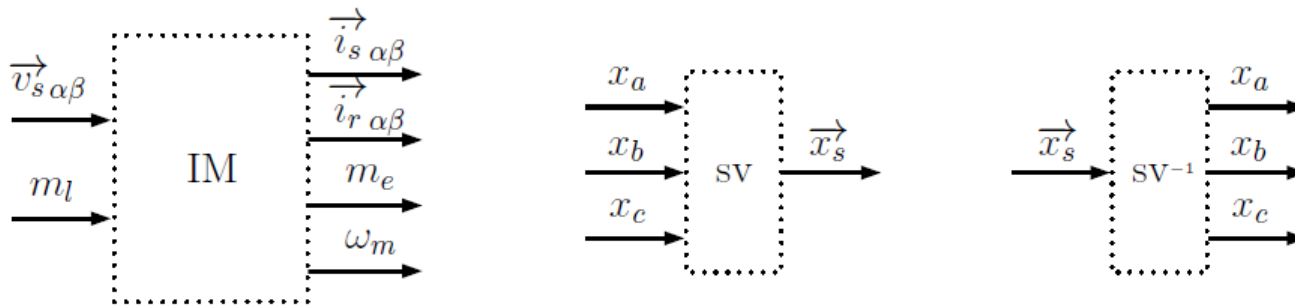
Simulink Model

Built a Simulink scheme of the previous IM model:

- create a subsystem for the space vector transformation
- create a IM subsystem in stationary $\alpha\beta$ components in sv form
- create a mask for input parameters

Hint:

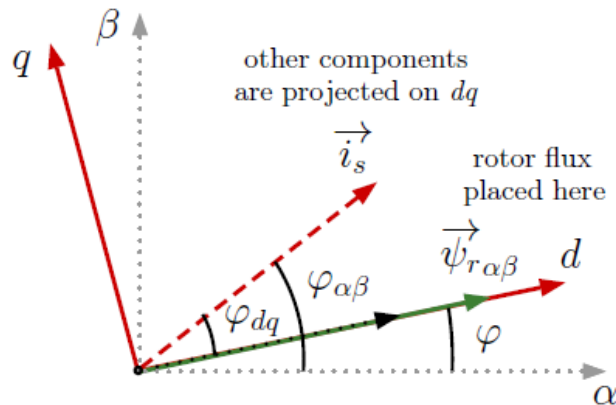
- realize the following blocks:



Rotor Field-Oriented Control (R-FOC)

Working principle

- Selecting an *appropriate* reference frame, the ***d*-component** of the stator current i_{sd} can be used to control the **rotor flux magnitude** $\|\psi_r\|$, while the ***q*-component** i_{sq} to control the **torque** m_e



R-FOC control choices

- The *d*-axis is aligned with the rotor flux vector ψ_r
 $\psi_{rd} = \|\vec{\psi}_{r\alpha\beta}\|$ and $\psi_{rq} = 0$
- The reference frame rotates synchronously with the rotor flux

$$\omega_{rf} = \omega_s$$

- The control action works similarly to DC motor control
- But, in DC motor the two currents flow in two distinct windings, here i_{sd}, i_{sq} are the components of a unique single-phase system

Rotor Field-Oriented Control (R-FOC)

Working principle

- R-FOC needs **matrix notation** using $\omega_{rf} = \omega_s$ and $\psi_{rd} = \left\| \overrightarrow{\psi_{r\alpha\beta}} \right\| :$

Under R-FOC choices

$$\begin{aligned}v_{sd} &= R_{ks} i_{sd} + L_{ks} p i_{sd} - \frac{R_r}{L_m} \psi_{rd} - \omega_s L_{ks} i_{sq} \\v_{sq} &= R_{ks} i_{sq} + L_{ks} p i_{sq} + \omega_m \psi_{rd} + \omega_s L_{ks} i_{sd} \\p \psi_{rd} &= R_r i_{sd} - \frac{R_r}{L_m} \psi_{rd} \quad (\star^1) \\0 &= R_r i_{sq} - (\omega_s - \omega_r) \psi_{rd} \\m_e &= n_p \psi_{rd} i_{sq} \quad (\star^2)\end{aligned}$$

Canonical control form

$$\begin{aligned}p i_{sd} &= \frac{1}{L_{ks}} \left[v_{sd} - R_{ks} i_{sd} + \frac{R_r}{L_m} \psi_{rd} + \omega_s L_{ks} i_{sq} \right] \\p i_{sq} &= \frac{1}{L_{ks}} \left[v_{sq} - R_{ks} i_{sq} - \omega_m \psi_{rd} - \omega_s L_{ks} i_{sd} \right] \\p \psi_{rd} &= R_r i_{sd} - \frac{R_r}{L_m} \psi_{rd} \quad (\star^1) \\0 &= R_r i_{sq} - (\omega_s - \omega_r) \psi_{rd} \\m_e &= n_p \psi_{rd} i_{sq} \quad (\star^2)\end{aligned}$$

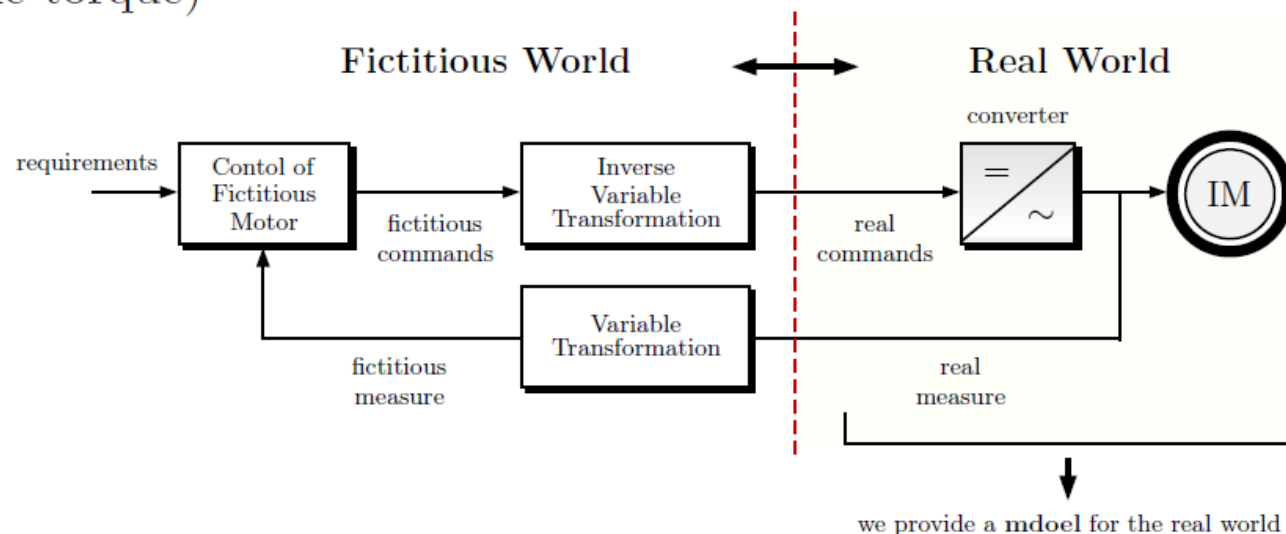
- $(\star^1)$ the **flux** depends only on i_{sd} $(\star^2)$ i_{sq} acts only on the **torque**
- If $\psi_{rd} = \text{cost}$ (properly use of iron), m_e changes according to i_{sq} whose reference can be directly derived from a torque reference m_e^*

The R-FOC of IM behaves similarly to DC motor control ☑

Aim of Vector Control Theory

R-FOC philosophy

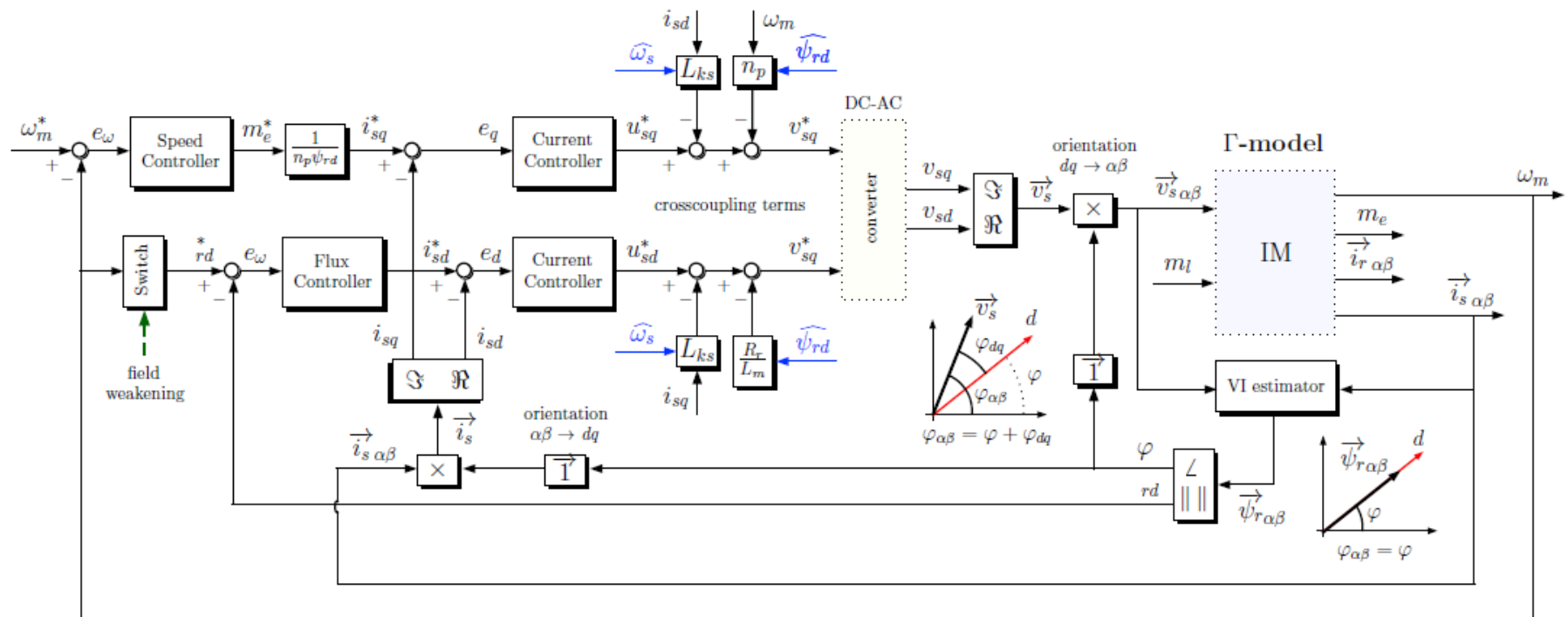
- The vector control theory tries to recreate on AC motor the electromechanical behavior of DC motor, by introducing the control of a fictitious motor, equivalent to the original one through appropriate mathematical transformations.
- Through R-FOC we obtain two decoupled dynamics for *slower transient* (i_{sd} related to the rotor flux) and *faster transient* effects (i_{sq} related to the torque)



Direct R-FOC of an IM – Simulink Scheme

Functional scheme

- We have to build this complete scheme:



Remark We must be careful on field coordinates *orientation*

Decoupled Current Controller dq

Current control loop and feedforward actions

- Let's consider:
$$pi_{sd} = \frac{1}{L_{ks}} \left[v_{sd} - R_{ks} i_{sd} + \underbrace{\frac{R_r}{L_m} \psi_{rd} + \omega_s L_{ks} i_{sq}} \right]$$

I can compensate them
↓
adding feedforward terms

- We add feedforward terms to the PIs output, allowing **decoupled** d -current control and q -current control during steady-state (only!)
- This trick lead also to design both the current control loops on the transfer function:

$$pi_{sx} = \frac{1}{L_{ks}} (v_{sx} - R_{ks} i_{sx}) \rightarrow$$

dq -Current Control

$$G_{ix}(s) = \frac{i_{sx}}{v_{sx}} = \frac{1}{R_{ks} + sL_{ks}} \quad x = d, q$$

This is equivalent to e compensation in DC motor

Cascade Control Approach

Speed control and flux control

- As already explained, if the inner loop is faster enough compared to the outer one we can *neglect* his effect in the outer design
- We have two outer loop (speed, flux), both closed on the mechanical speed ω_m :

Speed Control Loop

$$J \frac{d\omega_m}{dt} + \beta \omega_m = m_e - \cancel{m_l} \rightarrow$$

Speed Control

$$G_\omega(s) = \frac{\omega_m}{m_e} = \frac{1}{\beta + sJ}$$

Flux Control Loop

$$(\star^1) \quad p\psi_{rd} = R_r i_{sd} - \frac{R_r}{L_m} \psi_{rd} \rightarrow$$

Flux Control

$$G_{\psi_r}(s) = \frac{\psi_{rd}}{i_{sd}} = \frac{L_m}{1 + s \frac{L_m}{R_R}}$$

- Now we can design the PI controllers $(kp + ki/s)$ independently....

VI Estimator

Rotor flux estimation (alfa-beta)

Direct vs Indirect R-FOC

In brief, a Direct R-FOC use an estimator to provide the angular position φ from v_s, i_s , while an Indirect method use an encoder to measure φ .

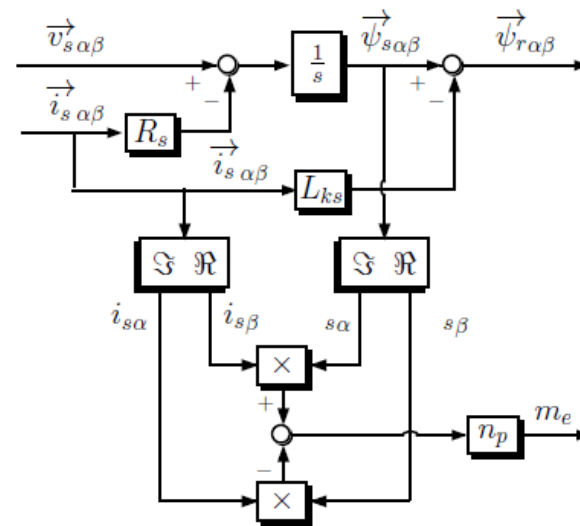
- A **VI estimator** is based on v_s and i_s measurement, referred to axes fixed with stator $\alpha\beta$, in order to estimate ψ_s, ψ_r and m_e .
- Rewrite the Γ -model equations:

$$\vec{\psi}_{s\alpha\beta} = \int (\vec{v}_{s\alpha\beta} - R_s \vec{i}_{s\alpha\beta}) dt$$

$$\vec{\psi}_{r\alpha\beta} = \vec{\psi}_{s\alpha\beta} - L_{ks} \vec{i}_{s\alpha\beta}$$

$$m_e = n_p (\psi_{s\alpha} i_{s\beta} - \psi_{s\beta} i_{s\alpha})$$

- A low pass filter is used instead $\frac{1}{s}$



Field Weakening Management

Working above the base speed

$$0 = R_r i_{sq} - (\omega_s - \omega_r) \psi_{rd} \quad \text{and} \quad p \psi_{rd} = R_r i_{sd} - \frac{R_r}{L_m} \psi_{rd}$$

we are going to exploit the operating region behavior as:

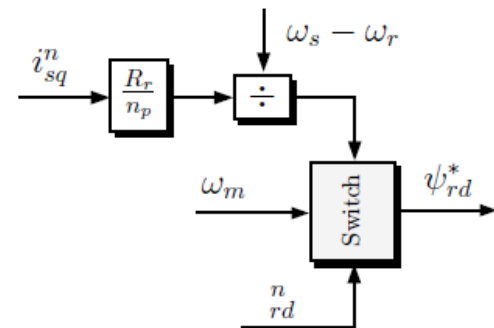
- if $\omega_m < \omega_b$ (*proper use of the iron*)

$$\psi_{rd}^* = \psi_{rd}^n = \text{cost}$$

(mathematically $\cancel{p \psi_{rd}^n} = R_r i_{sd}^n - \frac{R_r}{L_m} \psi_{rd}^n \rightarrow \psi_{rd}^n = L_m i_{sd}^n$)

- if $\omega_m > \omega_b$

$$i_{sq} = i_{sq}^n = \text{cost} \quad \rightarrow \quad \psi_{rd}^* = \frac{R_r i_{sq}^n}{\omega_s - n_p \omega_m}$$



Exercise

Electrical drive for an air compressor

Consider the air treatment system of a blast furnace based on a centrifugal compressor (CC) driven by an induction motor with the following characteristics:

- rated voltage 380 V
- rated frequency 50 Hz with pole pairs $n_p = 3$
- stator resistance $R_s = 0.24\Omega$ and inductance $L_s = 59.4\text{ mH}$
- rotor resistance $R_r = 0.175\Omega$ and inductance $L_r = 59.1\text{ mH}$
- mutual inductance $L_m = 57\text{ mH}$

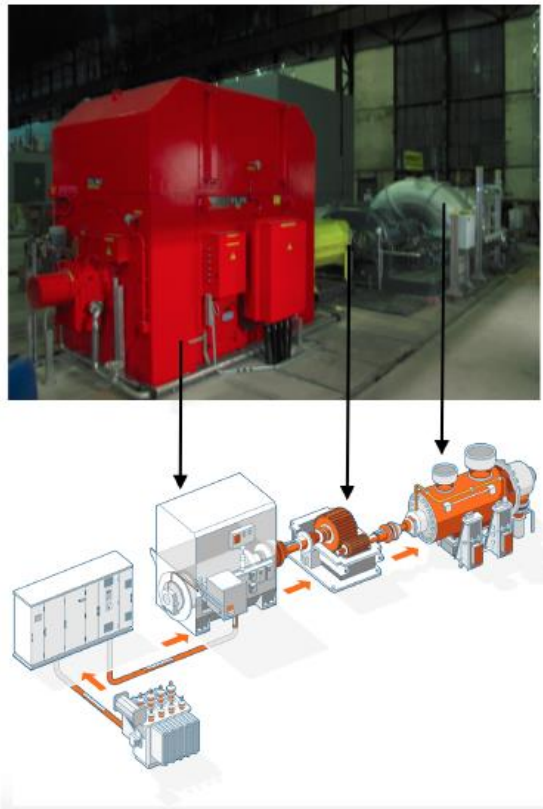
The compressor is coupled to the IM with a gearbox $r = \omega_l / \omega_m = 4$:

- the equivalent inertia seen from the motor side is $J_{eq} = 0.4\text{ kgm}^2$
- friction coefficient $\beta = 0.068$

The compressor behaviour see a load torque proportional to the square of the shaft speed as $m_l = k\omega_m^2$ where $k = 0.009\text{ Nm/rad}^2/\text{s}^2$

Exercise

Electrical drive for an air compressor



Industrial Project

- Simulate the IM without load at nominal voltage in order to evaluate the starting transient by figuring out:
 - stator i_s and rotor current i_r
 - torque m_e vs speed ω_m
 - position of ψ_s and ψ_r in $\alpha\beta$ reference frame
- Design and simulate speed, current and flux control in order to cover the given fluid demand at least one decade faster than the intrinsic characteristic of the uncontrolled system. Figure out:
 - currents, torque, speed profiles compared to the reference one

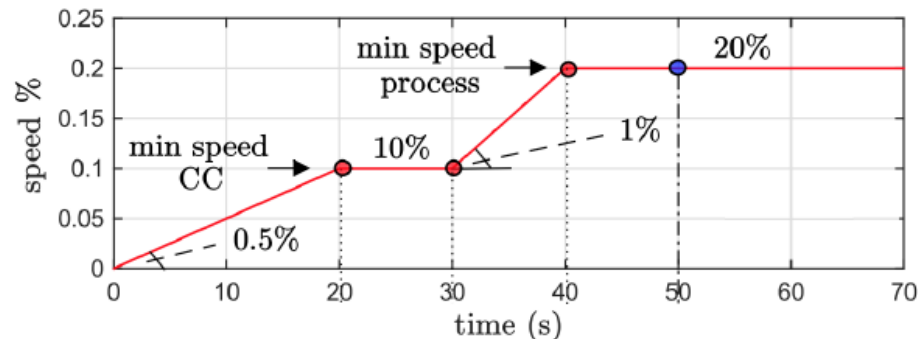
Simulation Hint

Electrical drive for an air compressor

up to you - *be careful*

Use the IM model equations to compute the **stator angular speed** ω_s in order to properly add the cross-coupling terms, realizing a real decoupled current control in dq

- The reference speed profile to follow is:
(percentage of the rated speed in **rpm** - *giri/min*)



- Select a proper bandwidth for the two control loop able to present phase margin $\varphi_m \geq 60^\circ$ and cascade constraints

Operating Procedure

To do list for a control problem in electrical drives field

- ① Modeling of v, i, φ
- ② Reduce problem complexity
- ③ Choose the point of view $\rightarrow$ angle $\rightarrow$ reference frame
- ④ Time constant and time constraints
- ⑤ Energy Balance $\rightarrow$ torque expression
- ⑥ Control schemes:
 - ① speed/position control
 - ② torque control
 - ③ flux control
- ⑦ Estimators/sensors
- ⑧ Converter modeling and power supply
- ⑨ Limits of operations